BT402 - FINALTERM SUBJECTIVE MEGA FILE

ALL DATA FROM PAST PAPERS

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Current papers

1. Why we use biodiesel?

Biodiesel is a domestically produced, clean-burning, renewable substitute for petroleum diesel. Using biodiesel as a vehicle fuel increases energy security, improves air quality and the environment, and provides safety benefits.

2. Define compost

Compost is organic matter that has been decomposed and recycled as a fertilizer and soil amendment. Compost is a key ingredient in organic farming. End product of the decomposition of organic matter

3. Define Vaccine. 2 Marks

A vaccine is a biological preparation that improves immunity to a particular disease. A vaccine typically contains an agent that resembles a disease-causing microorganism and is often made from weakened or killed forms of the microbe.

4. Advantage of biofertilizer?

The biofertilizers has special contribution to agriculture due to the following advantages:

- Biofertilizers act as supplements to chemical fertilisers.
- Biofertilizers are cost-friendly and can aid to decrease consumption of such fertilisers.
- Microbes in biofertilizers provide atmospheric nitrogen directly to plants.
- They aid in solubilisation and mineralisation of other plant nutrients like phosphates.
- Better synthesis and availability of hormones, vitamins, auxins and other growth-promoting substances improves plant growth.

- On an average crop yield elevates by 10–20 percent by their use.

5. What is probiotics

Probiotics are live microorganisms promoted with claims that they provide health benefits when consumed, generally by improving or restoring the gut microbiota. Probiotics are considered generally safe to consume, but may cause bacteria-host interactions and unwanted side effects in rare cases. There is some evidence that probiotics are beneficial for some conditions, but there is little evidence for many of the health benefits claimed for them.

6. Name biopesticdes uses in plants animals and microorganisms?

Biopesticides are certain types of pesticides derived from such natural materials as animals, plants, bacteria, and certain minerals. **For example**, canola oil and baking soda have pesticidal applications and are considered biopesticides.

7. Genetically engineered bacteria

Some bacteria have been genetically engineered

Why?

• To produce certain vaccines that are used to prevent infectious diseases

• Insulin, produced to treat diabetes

8. What is Matrix Assisted Laser Desorption/Ionization?(2)

MALDI is the abbreviation for "Matrix Assisted Laser Desorption/Ionization. " The sample for MALDI is uniformly mixed in a large quantity of matrix. The matrix absorbs the ultraviolet light (nitrogen laser light, wavelength 337 nm) and converts it to heat energy. A small part of the matrix (down to 100 nm from the top outer surface of the Analyte in the diagram) heats rapidly (in several nano seconds) and is vaporized, together with the sample.

9. Write the name of vitamins that are produce with microbes?

Vitamins are some organic compounds which are capable of performing many lifesustaining functions inside our body. • These compounds cannot be synthesised by humans, and therefore, they have to be supplied in small amounts in the diet.

• Microbes are capable of synthesizing these compounds and hence they can be used for the commercial production of many of the Vitamins e.g. thiamine (Vitamin B1), riboflavin (Vitamin B2)

10. Define Subunit Vaccine and give example. 3 Marks

- Instead of the entire microbe, subunit vaccines include only the antigens that best stimulate the immune system.

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- In some cases, these vaccines use epitopes the very specific parts of the antigen that antibodies or T cells recognize and bind to immunization.
- Because subunit vaccines contain only the essential antigens and not all the other molecules that make up the microbe.

Example: Plague immunization.

11. Types of antimicrobial 5 marks

Infections and diseases may be caused by different types of organisms like bacteria, fungi, and viruses, etc., in humans and animals. The drug used to prevent the pathogenicity of microorganisms is called an antimicrobial agent.

Examples: Antibiotics, antiseptics, and disinfectants. Let us look at some antimicrobial agents and their types with examples.

Antimicrobial agents are used to preventing infections and diseases caused by pathogens. Different types of antimicrobial drugs are commonly available. These are as follows:

Antibacterial drug: A drug that is used to inhibit the pathogenic activity of bacteria is called an antibacterial drug. Example: Zithromax.

Antifungal drug: A drug that is used to prevent fungal activity in the host is called an antifungal drug. Example: Miconazole

Antiviral agent: A drug which is used to stop the pathogenic action of a virus is called antiviral agents. Example: Tamiflu.

Antiparasitic drug: A drug that is used to prevent the growth of pathogenic parasites. Example: Anthelmintics

12. Possible side effect of prebiotic

Most prebiotics and probiotics can be safely consumed by most healthy adults without side effects. In some cases, while your digestive system adjusts, abdominal discomfort, bloating and gas may occur.

13. Biofuels

• A biofuel is defined as any fuel whose energy is obtained through a process of biological carbon fixation.

• Biofuels are fuels produced directly or indirectly from organic material biomass, including plant materials and animal waste.

14. Three names of fungi used in food and beverage processing (2)

Aspergillus oryzae: Fungus involved in the making of soy sauce and miso.

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Penicillium camemberti: Species of fungus used in the making of Camembert Cheese and other related cheese.

Penicillium roqueforti: Species of fungus used in the making of Roquefort Cheese, as well as other blue cheeses.

15. Role of Biotechnology in agriculture (3)

Biotechnology provides farmers with tools that can make production cheaper and more manageable. For example, some biotechnology crops can be engineered to tolerate specific herbicides, which make weed control simpler and more efficient. Other crops have been engineered to be resistant to specific plant diseases and insect pests, which can make pest control more reliable and effective, and/or can decrease the use of synthetic pesticides. These crop production options can help countries keep pace with demands for food while reducing production costs.

16. DNA vaccine with example (3)

DNA vaccine

A DNA vaccine is a type of vaccine that transfects a specific antigen-coding DNA sequence into the cells of an organism as a mechanism to induce an immune response.

- Still in the experimental stages, these vaccines show great promise, and several types are being tested in humans
- DNA vaccines take immunization to a new technological level
- These vaccines dispense with both the whole organism and its parts and get right down to the essentials: the microbe's genetic material.

Example: Influenza vaccine.

17. Two types of probiotics

Bifidobacteria: This genus of bacteria is commonly used in foods and supplements. They're thought to:

- support the immune system
- limit the growth of harmful bacteria in the intestine
- help break down lactose into nutrients the body can use

Lactobacillus: This genus of bacteria produces lactase, the enzyme that breaks down lactose, or milk sugar. These bacteria also produce lactic acid. Lactic acid helps control the population of bad bacteria. It also serves as muscle fuel and increases the body's absorption of minerals. Lactobacillus bacteria are found naturally in the:

- mouth
- small intestine

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- vagina

18. Why pants use nitrates?2 mark

They need nitrates to make amino acids. Pea and bean plants use bacteria Rhizobium which directly convert nitrogen into acid amino

19. Define Soil Treatment? 2 Marks

Soil treatment

All the bio fertilizers along with the compost fertilizers are mixed together. They are kept for one night. Then the next day the mixture is spread on the soil where seeds have to be sown.

20. What is lactic acid fermentation? (3 marks) (topic 25)

Lactic acid fermentation causes you to feel a burning sensation in your working muscles during exercise but not after. ... Your body at work uses muscles, which require constant fueling of oxygen in a process called aerobic metabolism where energy is generated from the muscles using oxygen.

21. define biopesticides and biofertilizers

A **Biopesticide** is a biological substance or organism that damages, kills, or repels organisms seens as pests. Biological pest management intervention involves predatory, parasitic, or chemical relationships. They are obtained from organisms including plants, bacteria and other microbes, fungi, nematodes, etc.

A **biofertilizer** is a substance which contains living micro-organisms which, when applied to seeds, plant surfaces, or soil, colonize the rhizosphere or the interior of the plant and promotes growth by increasing the supply or availability of primary nutrients to the host plant

22. Economic significance of microbial biotechnology

Biotechnology is generally defined as application of living systems based technologies to develop commercial processes and products. Over the last few decades, several fundamental discoveries in life sciences have given rise to Modern Biotechnology which is now one of the fastest growing areas of science; hence this century has rightly been termed as ‘Century of Biology’, hoping that such advances in life sciences will yield changes more momentous than those of electricity and computers. In view of these developments, Biotechnology was included among the six priority areas of Science & Technology by the National Commission of Science & Technology.

Biology in this century has become an information science. Many programs and initiatives underway at major research institutions and leading companies are already giving shape to this assertion.

23. Define Live attenuated vaccines and give example? 3 Marks

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Live attenuated vaccines contain a version of the living microbe that has been weakened in the lab so it can't cause disease. Because a live, attenuated vaccine is the closest thing to a natural infection, these vaccines are good "teachers" of the immune system.

Example: Vaccines against measles, mumps, and chickenpox.

24. Write down the name of the steps in the production of vaccine? 5 Marks

The production of a vaccine can be divided in the following steps:

1. Generation of the antigen

The first step in order to produce a vaccine is generating the antigen that will trigger the immune response. For this purpose the pathogen's proteins or DNA need to be grown and harvested using the following mechanisms:

Viruses are grown on primary cells such as cells from chicken embryos or using fertilised eggs (e.g. influenza vaccine) or cell lines that reproduce repeatedly (e.g. hepatitis A)

Bacteria are grown in bioreactors which are devices that use a particular growth medium that optimizes the production of the antigens

Recombinant proteins derived from the pathogen can be generated either in yeast, bacteria or cell cultures.

2. Release and isolation of the antigen

The aim of this second step is to release as much virus or bacteria as possible. To achieve this, the antigen will be separated from the cells and isolated from the proteins and other parts of the growth medium that are still present.

25. how microbes can influence and make evolution of oil recovery

Many experts believe that practical breakthroughs in these technologies are decades away at best. So, the challenge is to find a fuel that can replace the practical qualities of oil (like being easy to drive around), but which does not pollute the same way. The solution, atleast for now, appears to be algal-based biofuels, which are still years if not decades away from commercialization. The idea is simple. Algae have lipid and lipid can be converted to a number of fuels including diesel, ethanol, butanol, and methanol. Because algae absorb CO2 to make lipid, the net impact on the environment should be very small.

26. how prebiotics are produced

Prebiotics are indigestible oligosaccharides commonly produced from food sources such as chicory root, milk, and oats , and are known to promote the growth of beneficial probiotic microorganisms in the lower intestine of mammals.

27. Bio fertilizers

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A **biofertilizer** is a substance which contains living micro-organisms which, when applied to seeds, plant surfaces, or soil, colonize the rhizosphere or the interior of the plant and promotes growth by increasing the supply or availability of primary nutrients to the host plant

28. Vitamin and name of important vitamins

Vitamins are some organic compounds which are capable of performing many life-sustaining functions inside our body. These compounds cannot be synthesized by humans, and therefore, they have to be supplied in small amounts in the diet. Microbes are capable of synthesizing these compounds and hence they can be used for the commercial production of many of the Vitamins e.g. thiamine (Vitamin B1), riboflavin (Vitamin B2), pyridoxine (Vitamin B6), folic acid, pantothenic acid (Vitamin B5), biotin (Vitamin B7), Vitamin B12, ascorbic acid (Vitamin C).

29. Antivirals.

Antiviral drugs are a class of medication used specifically for treating viral infections. Like antibiotics, specific antivirals are used for specific viruses. They are relatively harmless to the host and therefore can be used to treat infections. They should be distinguished from viricides, which actively deactivate virus particles outside the body. Many antiviral drugs are designed to treat infections by retroviruses, mostly HIV. Important antiretroviral drugs include the class of protease inhibitors. Herpes viruses, best known for causing cold sores and genital herpes, are usually treated with the nucleoside analogue acyclovir.

30. Importance of microbial

Biotechnology is generally defined as application of living systems based technologies to develop commercial processes and products. Over the last few decades, several fundamental discoveries in life sciences have given rise to Modern Biotechnology which is now one of the fastest growing areas of science; hence this century has rightly been termed as ‘Century of Biology’, hoping that such advances in life sciences will yield changes more momentous than those of electricity and computers. In view of these developments, Biotechnology was included among the six priority areas of Science & Technology by the National Commission of Science & Technology.

31. It is possible to make protein array?

Indeed this is possible. In one study, 5800 open reading frames in the yeast genome were cloned and expressed as a fusion with GST and a hexa histidine tag. The proteins were purified on glutathione-agarose and attached to glass slides coated with nickel, using the affinity of the hexa histidine tag. The utility of such a “chip” was demonstrated when it was treated with a solution of biotinylated calmodulin. When the binding of the biotin tag was detected with a fluorescence-labeled streptavidin, 39 proteins were shown to bind calmodulin, out of which only six were the already known calmodulin binding proteins (Figure 4.6). Similar approaches should be possible with many reagents, including drug candidates, and protein arrays may one day become the standard approach for searching for proteins with specific functions on a global scale.

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32. Why studying microbe is very important

There are pros and cons to everything we know, and the same applies to these microorganisms; some microbes are deadly to the planet's health while others help keep this planet healthy.

For example, studying microbes is essential in the medical sector to understand how these tiny organisms cause diseases and their diagnosis and cure. Further, understanding these organisms also led to discovering their uses for human health as nutrition or medicine and curating innovative agricultural products.

33. Define biotechnology?

Biotechnology is the integration of natural sciences and engineering sciences in order to achieve the application of organisms, cells, parts thereof and molecular analogues for products and services. The term biotechnology was first used by Károly Ereky in 1919, meaning the production of products from raw materials with the aid of living organisms.

34. How animals digest cellulose?

The rumen is the first compartment where ingested food containing cellulose is stored temporarily and later regurgitated to chew their cud. They are able to digest cellulose because of the presence of bacteria and enzymes in the rumen where anaerobic bacterial digestion occurs.

35. Key roles of adjuvants?

An adjuvant is a substance that is formulated as part of a vaccine to enhance its ability to induce protection against infection. The word "adjuvant" comes from the Latin adjuvare and means "to help." Adjuvants help activate the immune system, allowing the antigens—pathogen components that elicit an immune response—in vaccines to induce long-term protective immunity

36. Any two challenges of adjuvants? (find answer)

37. What is the basic technique use for the sequence earlier? (find answer)

38. What is nitrogen biofertilizers?

A biofertilizer is a substance which contains living micro-organisms which, when applied to seeds, plant surfaces, or soil, colonize the rhizosphere or the interior of the plant and promotes growth by increasing the supply or availability of primary nutrients to the host plant.[1] Biofertilizers add nutrients through the natural processes of nitrogen fixation, solubilizing phosphorus, and stimulating plant growth through the synthesis of growth-promoting substances

39. Define metabolomics?

Metabolomics is the large-scale study of small molecules, commonly known as metabolites, within cells, biofluids, tissues or organisms. Collectively, these small molecules and their interactions within a biological system are known as the metabolome.

40. How biofuel generate electricity 3 marks

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Charging Electronics

- According to scientists from Saint Luis University, a fuel cell was developed with cooking oil and sugar to generate electricity.
- Consumers will be able to use these cells instead of generating electricity.
- Consumers may be able to use fuel cells in place of batteries to charge anything from computers to cell phones.

41. Microarray

A **microarray** is a laboratory tool used to detect the expression of thousands of genes at the same time. DNA microarrays are microscope slides that are printed with thousands of tiny spots in defined positions, with each spot containing a known DNA sequence or gene. Often, these slides are referred to as gene chips or DNA chips. The DNA molecules attached to each slide act as probes to detect gene expression, which is also known as the transcriptome or the set of messenger RNA (mRNA) transcripts expressed by a group of genes

42. Generation of antigen virus bacteria for vaccine production

The first step in order to produce a vaccine is generating the antigen that will trigger the immune response. For this purpose the pathogen's proteins or DNA need to be grown and harvested using the following mechanisms. For this the virus or microbe is grown either on primary cells such as chicken eggs (e.g. in influenza) or on cell lines or cultured human cells (e.g. Hepatitis A). Bacteria against which the vaccines are developed may be grown in bioreactors (e.g. Haemophilus influenzae type b).

The antigen may also be a toxin or toxoid from the organism (e.g. Diphtheria or tetanus) or it may be part of the microorganism as well. Proteins or parts from the organism can be generated in yeast, bacteria, or cell cultures. Bacteria or viruses may be weakened using chemicals or heat to make the vaccine (e.g. polio vaccine).

43. three bacteria for food and beverage

- Campylobacter.
- Clostridium perfringens.
- E. coli.
- Listeria

44. uses of antimicrobial products

- Triclosan, an antimicrobial compound, has been found to promote oral health and aid in the fight against gingivitis when added to toothpaste.

- Kitchen counters, office desks, bathroom sinks, and other high-traffic areas in homes and offices may accumulate germs that can make people sick. Cleaning products without antibacterial/antimicrobial ingredients will clean the surface, but will not kill the germs on it.

45. Bt cotton in pakistan

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Pakistan is the world's fourth largest producer of cotton after China, the USA and India, according to statistics from the All Pakistan Textile Mills Association. Cotton and textiles make up over 60 per cent of Pakistan's Rs. 488.00 billion annual export. Cotton or white gold as it is aptly called is grown for its lint and seed, which yield cotton fiber and seed oil, respectively. This crop occupies 70-75 millions acre of world area with a production of 20-25 metric tons. In Pakistan its area spans over 12-14 million acre with an average yield of 485kg/acre or 210kg/hectare of lint and 500kg/acre of seed cotton.

By the introduction of the Bt cotton in Pakistan could result in a 45-55 per cent reduction in insecticide use on cotton (Which is 85per cent of Rs11 billion). This would mean a benefit of about of about Rs 4.2 to Rs 5.40 billion apart from the favorable impact on the environment and increase in cotton yield.

In Pakistan, average yield of conventional cotton per acre is around 25-28 maund or 933 kg-1,044 kg. Bt Cotton in Pakistan can increase per acre yield from 14 to 30 per cent. Which means that, on the one hand, it will bring prosperity for Pakistani farmers, on the other, it will bring a boom to all industries and business activities which are directly or indirectly associated with agriculture sector.

46. Polyester

Polyester is a synthetic fabric that's usually derived from petroleum. This fabric is one of the world's most popular textiles, and it is used in thousands of different consumer and industrial applications.

Chemically, polyester is a polymer primarily composed of compounds within the ester functional group. Most synthetic and some plant-based polyester fibers are made from ethylene, which is a constituent of petroleum that can also be derived from other sources.

47. Metagenomics

A fundamental operation in biotechnology is the expression of a foreign gene coding for a useful production a suitable host organism, such as E. coli or yeast. However, because what is cloned is a piece of DNA, it may come directly from environmental sources. This approach of direct cloning is extremely valuable because most of the microorganisms in the environment have not yet been cultured, and the use of samples (for example soil samples) from a diverse range of environments is likely to supply genes that code for proteins with widely different properties. Thus, it has now become a standard approach in the industry to maintain such samples for the direct cloning of useful genes.

We can extend this approach, and instead of cloning just a gene or a gene cluster, we may hope to reconstruct the entire genomes of yet-to-be-cultured organisms from DNA samples in the environment. We will then be dealing not with the genome of a single cultured organism, but with genomes of the entire community of microorganisms, many of which may be currently uncultured. This is what is often called **metagenomics.**

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48. Different btw homology and paralogs?

homology, in biology, similarity of the structure, physiology, or development of different species of organisms based upon their descent from a common evolutionary ancestor. Homology is contrasted with analogy, which is a functional similarity of structure based not upon common evolutionary origins but upon mere similarity of use.

Homologous proteins that exist in the same species, presumably created originally by gene duplication and often differentiated later in terms of functions, are called **paralogs.**

49. Sequencing history

The first major breakthrough in sequencing technology was made by Fredrick Sanger in 1977, when he and his colleagues introduced the "dideoxy" chain-termination method for sequencing DNA molecules, also known as "Sanger Sequencing".

It earned him his second Nobel Prize. This method was later used by Sanger and colleagues to sequence human mitochondrial DNA (16,569 base pairs), and bacteriophage λ (48,502 base pairs) – the first complete genome. Sanger sequencing was the most widely used form of sequencing for over 30 years.

50. types of composting

- Composting Basics.
- Onsite Composting.
- Vermicomposting.

51. PIP

Plant-Incorporated-Protectants (PIPs) are pesticidal substances that plants produce from genetic material that has been added to the plant. For example, scientists can take the gene for the Bt pesticidal protein and introduce the gene into the plant's own genetic material. Then the plant, instead of the Bt bacterium, manufactures the substance that destroys the pest. The protein and its genetic material, but not the plant itself, are regulated by EPA.

52. Write note on 2nd and 3rd generation of biofuel?

Second-generation

- Second-generation biofuels are produced from non-food crops, such as cellulosic biofuels and waste biomass (stalks of wheat and corn, and wood).
- Common second-generation biofuels include vegetable oils, biodiesel, bio alcohols, biogas, solid biofuels, and syngas.
- Research continues on second-generation biofuels including bio hydrogen, bio methanol, DMF, Bio-DME, Fischer-Tropsch diesel, bio hydrogen diesel, mixed alcohols and wood diesel.

53. Bio butanol

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The fuel is produced through fermentation of corn feedstock and the process is nearly identical to fuel ethanol production from corn. In addition to transportation fuel, biobutanol companies produce a range of coproducts, such as solvents/coatings, plastics, and fibers.

54. Main objectives of transgenic plants.

The main purpose in the production of transgenic plants is to produce crops, which have ideal traits, quality, and high yield. Besides being beneficial to the agriculture sector, the plants are found to be able to act as the factory for pharmaceutical protein production

55. Important role of Biofertilizers

Biofertilizers are required to restore the fertility of the soil. Prolonged use of chemical fertilizers degrades the soil and affects crop yield. Biofertilizers, on the other hand, enhance the water holding capacity of the soil and add essential nutrients such as nitrogen, vitamins and proteins to the soil. They are the natural form of fertilizers and hence, widely used in agriculture.

56. Importance of microorganisms in plants.

Microbes are the key components of soil nutrient cycling. Status of soil health and richness of soil nutrient pool depend on structure and functions of soil microbial community. Microbes play an important role in nutrient mobilisation and uptake.

They promote plant growth and suppress disease by their various activities. Phosphate and sulphate solubilisation, plant growth promotion, siderophore production, nitrogen fixation, denitrification, immune modulation, signal transduction and pathogen control are some of the well-recognised microbial mediated processes which promote the plant growth and protect them from pests.

57. define toxoid vaccine

For bacteria that secrete toxins, or harmful chemicals, a toxoid vaccine might be the answer. These vaccines are used when a bacterial toxin is the main cause of illness. Scientists have found that they can inactivate toxins by treating them with formalin, a solution of formaldehyde and sterilized water. Such “detoxified” toxins, called toxoids, are safe for use in vaccines. When the immune system receives a vaccine containing a harmless toxoid, it learns how to fight off the natural toxin. The immune system produces antibodies that lock onto and block the toxin. Vaccines against diphtheria and tetanus are examples of toxoid vaccines.

58. How microbes helpful in plant growth?

Microbes improve plant growth by enhancing the availability of nutrients, the regulation of phytohormones, and increasing plant tolerance against stresses. PGPM act as biofertilizer, increasing macro and micronutrient availability.

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They increase the concentrations of auxin, gibberellin, cytokinin, ACC-deaminase, and reduce ethylene levels. Beneficial microbes also produce volatile metabolites (VOC), that may induce disease resistance and abiotic stress tolerance.

59. Two types of bio fertilizers

Nitrogen bio fertilizers

The sort of bio fertilizers can help any farmers to determine the nitrogen point during the solid ground. Nitrogen may be a crucial factor which happens to be put to use in any growing for put. Including, Azotobacteria must be used to your non-legume bounty; Rhizobium it takes to your legume bounty. Purple earth-friendly algae are needed to progress hemp despite the fact that Acetobacter must be used to progress sugarcane.

Phosphorus bio fertilizers

Phosphorus bio fertilizers widely used to determine the phosphorus point during the solid ground. The decision for phosphorus to your put growing is small. Phosphorus bio fertilizers come up with any solid ground obtain demanded sum of phosphorus. Isn't crucial that your particular selected phosphorus bio fertilizers must be used for that selected types of head. They are put to use in all different kinds of any head including; Acetobacter, Rhizobium can implement.

60. Prokaryotes genome detail

There is a large variation in the size of bacterial genomes. Compared with the H. influenzae genome (1.8 Mb), some obligate parasites, such as Chlamydia trachomatis (1.0 Mb) and Mycoplasma genitalum (0.6 Mb), have smaller genomes. On the other hand, the genome of E. coli (4.6 Mb) is much larger, and even larger genomes are found in Pseudomonas aeruginosa (6.3 Mb), Streptomyces coelicolor (8.7 Mb), and Bradyrhizobium japonicum (9.1 Mb). (With rare exceptions, most archeal genomes have a rather small size, between 1.5 and 3 Mb). In the prokaryotic genome, most of the DNA represents coding sequences or genes, unlike in higher animals and plants. So this large difference leads us to two questions. The first is, what is the minimal set of genes that allow simple cellular organism to grow and replicate? Here we have to classify genes, and the proteins coded by these genes, on the basis of homology.

Homology means descent from a common ancestor, and this is usually predicted by comparing sequences using computer programs such as BLAST. We distinguish two kinds of homologs. Homologous proteins in two different organisms, where they developed independently of each other, are called orthologs. In contrast, homologous proteins that exist in the same species, presumably created originally by gene duplication and often differentiated later in terms of functions, are called paralogs.

61. viruses genome (long)

Viruses are small submicroscopic, obligate intracellular parasites, which contains either DNA or RNA as genome protected by a virus-encoded protein coat called capsid. Viruses are mobile

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genetic elements, depends on metabolic and biosynthetic machinery of host cells for their propagation. Viruses cannot carry out their life sustaining functions outside the host cell. They cannot synthesize proteins as they lack ribosomes and uses the host cell ribosome machinery for translating their mRNA into proteins.

Viruses can neither generate nor store ATP but derive the necessary energy and other metabolic functions by parasitizing the host cell for the basic materials such as amino acids, nucleotides, and lipids. Even though the viruses are speculated as a form of proto-life, their inability to survive outside living organisms does not make them as living organisms in the strict sense and it is highly unlikely that they preceded cellular life during the evolution of the earth.

Some scientists even speculate that viruses have begun as rogue segments of genetic code which got adapted to a parasitic form of existence.

62. write three methods of applying biofertilizers

Application-Method of applying biofertilizers

The important method of biofertilizers applications are listed below:

Seedling root dip: This method is usually applicable for rice crops. The seedlings are sowed in the bed of water and kept for 8-10 hours.

Seed Treatment: The seeds are soaked in the mixture of nitrogen and phosphorus fertilizers. These seeds are then left to dry and are sown as soon as possible.

Soil Treatment: The biofertilizers along with the compost fertilizers are blended and kept for one night. This mixture is then scattered on the soil where the seeds have to be sown.

63. What is basic technique of genome sequencing

The techniques used included Northern blot, which involves annealing of fractionated mRNA with the cognate pieces of DNA, and reverse transcriptase– PCR, in which mRNA is used as a template to produce cDNA, which is then amplified by PCR.

64. Prokaryotes genome

In prokaryotes, the genome is composed of a single, double-stranded DNA molecule in the form of a loop or circle. The region in the cell containing this genetic material is called a nucleoid. Some prokaryotes also have smaller loops of DNA called plasmids that are not essential for normal growth.

Bacteria can exchange these plasmids with other bacteria, sometimes receiving beneficial new genes that the recipient can add to their chromosomal DNA. Antibiotic resistance is one trait that often spreads through a bacterial colony through plasmid exchange.

65. How microbes are beneficial for biofertilizer

The Importance Bio-Fertilizers and Its Advantages

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- The Biofertilizer Microbes are utilized for fixing the nutrients within the Soil, especially related to the insufficient Nitrogen.
- The Biofertilizer is the living organism and it helps in converting the composite organic material into the simple compounds. This makes it easier for the plants or trees to take them.
- The usage of Microorganisms as fertilizers helps in enhancing the growth of plants enormously and it also gives the richness back to the Soil.
- Instead of controlling the growth, the Microorganisms help in advancing the process of growth within plants especially related to the growth in new seed.

66. Inactivated Vaccines.

• Scientists produce inactivated vaccines by killing the disease-causing microbe with chemicals, heat, or radiation. Such vaccines are more stable and safer than live vaccines. Because dead microbes can't mutate back to their disease-causing state

Example: Vaccines against influenza, polio, hepatitis A, and rabies

67. Microbial pesticide

Microbial pesticides consist of a microorganism (e.g., a bacterium, fungus, virus or protozoan) as the active ingredient. Microbial pesticides can control many different kinds of pests, although each separate active ingredient is relatively specific for its target pest[s]. For example, there are fungi that control certain weeds and other fungi that kill specific insects.

The most widely used microbial pesticides are subspecies and strains of Bacillus thuringiensis, or Bt. Each strain of this bacterium produces a different mix of proteins and specifically kills one or a few related species of insect larvae. While some Bt ingredients control moth larvae found on plants, other Bt ingredients are specific for larvae of flies and mosquitoes. The target insect species are determined by whether the particular Bt produces a protein that can bind to a larval gut receptor, thereby causing the insect larvae to starve.

68. Symbiotsis

Symbiosis is any relationship between two or more biological species. Such relationships are usually long term and have a strong impact on the fitness of one or both organisms. Symbiotic relationships are categorized by the benefits and physical relationships experienced by each species.

69. Importance of microbes in medicine (2)

Microbes are also used in pharmaceutical industries for synthesis of chemical drugs, chemical compounds and other compounds. It also leads to discovery of cell mechanisms allows pharmacists to discover antimicrobial drugs that would prevent an escalating number of communicable diseases.

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70. Plant-microbe interaction(3)

Applications of plant-microbe interactions

Although the potential use of plant-microbe interactions has been recommended as a means to maximize the economic returns of agribusiness, still there are widespread concerns regarding the use of plant-microbe chemical signaling and their interactive coordination.

71. Advantage of biopesticide (5).

Advantages of Biopesticides

- Biopesticides are usually inherently less toxic than conventional pesticides.
- Biopesticides generally affect only the target pest and closely related organisms, in contrast to broad spectrum, conventional pesticides that may affect organisms as different as birds, insects and mammals.
- Biopesticides often are effective in very small quantities and often decompose quickly, resulting in lower exposures and largely avoiding the pollution problems caused by conventional pesticides.
- When used as a component of Integrated Pest Management (IPM) programs, biopesticides can greatly reduce the use of conventional pesticides, while crop yields remain high.
- To use biopesticides effectively (and safely), however, users need to know a great deal about managing pests and must carefully follow all label directions.

72. Difference between bioleaching and biomining

Biomining utilizes naturally occurring prokaryotic communities. Here, microorganisms are used to leach metals, principally copper but also nickel and zinc, from low-grade sulfide- and/or iron-containing ores.

Bioleaching is a process in mining and biohydrometallurgy (natural processes of interactions between microbes and minerals) that extracts valuable metals from a low-grade ore with the help of microorganisms such as bacteria or archaea.

73. Why it is necessary to understand molecular mechanism of regulatory process? (find answer)

74. Bioprocessing of microbes

Bioprocessing or biotechnology is used in the production of pharmaceuticals, foods, flavours, fuels, and chemicals with the aid of a biocatalyst such as an enzyme, microorganisms, plant cell, or animal cell in a bioreactor.

75. Difference between submerged and solid state fermentation.

Submerged Fermentations:

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Submerged Fermentations are those in which the nutrient substratum is liquid and the organism grows inside the substratum. The culture conditions are made uniform with the help of spargers and impeller blades. Most of the industrial fermentations are of this type. The substratum which is in a liquid state and such medium is also called as broth.

Solid Substrate/State Fermentation:

Solid state (substratum) fermentation (SSF) is generally defined as the growth of the microorganism on moist solid materials in the absence or near the absence of free water. In recent years SSF has shown much promise in the development of several bioprocesses and products, SSF has been ambiguously used as solid-state fermentation or solid-substrate fermentation.

However, it is proper to distinguish between two processes. Solid substrate fermentation should be used to define only those processes in which the substrate itself acts as carbon source occurring in absence or near absence of free water. On the other hand, the solid state fermentation is that fermentation which employs a natural substrate as above or an inert substrate used as solid support.

76. How Second-generation biofuel are produced? 3 Marks

Second-generation biofuels are produced from non-food crops, such as cellulosic biofuels and waste biomass (stalks of wheat and corn, and wood). Common second-generation biofuels include vegetable oils, biodiesel, bio alcohols, biogas, solid biofuels, and syngas.

Research continues on second-generation biofuels including bio hydrogen, biomethanol, DMF, Bio- DME, Fischer-Tropsch diesel, biohydrogen diesel, mixed alcohols and wood diesel.

77. How yeast causes the bread to rise?

This yeast ferments the carbohydrates in the dough to produce carbon dioxide. In this process, warm water brings the yeast cells to action. They begin to eat and digest the sugars in the bread and flour, which is when the carbon dioxide and small amounts of ethanol alcohol is released. The carbon dioxide produced causes the dough to rise when the bubbles become trapped in the bread dough, and creates the bread's soft texture. The yeast normally needs a few hours for this process to occur in order to sufficiently cause the dough to rise

78. What is Biodiesel and how it is produced? 5 Marks

Biodiesel is produced, mainly in the European Union, by combining vegetable oil or animal fat with an alcohol. Biodiesel can be blended with traditional diesel fuel or burned in its pure form in compression ignition engines. Its energy content is somewhat less than that of diesel (88 to 95%). Biodiesel can be derived from a wide range of oils, including rapeseed, soybean, palm, coconut or jatropha oils and therefore the resulting fuels can display a greater variety of physical properties than ethanol.

79. Describe the aspects of polysaccharide structure. 5 marks

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The glycosidic bond between two monosaccharides is formed between the hydroxyl group on the anomeric carbon of one monosaccharide and any hydroxyl group on the other monosaccharide. Hence, the formation of a disaccharide from two identical hexopyranose ring structures of the d-series may result in 11 different isomers. In eight of the isomers, the glycosidic linkage is between C-1, in either the α or β anomeric configuration, and C-2, C-3, C-4, or C-6 of the other pyranose residues. These are designated α-D- (1→2) linkages, α-D-(1→3), linkages, and so on, where α and β refer to the anomeric configuration at C-1.

The other three isomers are created by acetal formation between both C-1 atoms through the glycosidic oxygen atom in the α,α, α,β, or β,β configuration. A similar series of 11 isomers will result if the two identical hexopyranose residues are of the l-series. The number of possible isomers is even higher if furanose forms (five-membered rings) are included. When the two monosaccharides are not identical, the number of possible structures increases, because either carbohydrate residue can occupy the first or the second position – that is, can be the reducing or non-reducing residue. Each additional carbohydrate residue brings a large increase in the number of possible isomers. For complex polysaccharides, the number of theoretically possible structures is astronomical.

80. Biochemical pesticides 2

They are naturally occurring substance to control pest by non-toxic mechanisms. Biochemical pesticides include substances as insect sex pheromones that interfere with mating that attract insect pest to traps.

81. Name of Fungi Involved in food/Brewing industry.

There are various types of fungus used in the food industry which are:

Saccharomyces cerevisiae: It is also called Brewer's yeast. It is a single-celled fungus microorganism that is very important in baking, brewing, and winemaking for a long time. It is the microorganism responsible for the most common type of fermentation. It is used in the brewing of beer as it is fermenting yeast which causes the production of carbon dioxide. It is also used in baking as a leavening agent in bread and other baked products.

Penicillium roqueforti: It is a saprophytic fungus. The major uses of these fungi are in the production of Roquefort cheese, as antifungal, and as flavoring agents.

Streptococcus thermophilus and Lactobacillus bulgaricus: These two different types of lactic acid bacteria are used in the production of curd, and yogurt. They are used as fermentation agents that produce the lactic acid.

82. What are Metalobomics?

Metabolomics is the large-scale study of small molecules, commonly known as metabolites, within cells, biofluids, tissues or organisms. Collectively, these small molecules and their interactions within a biological system are known as the metabolome.

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83. Benefits of Probiotics.

Probiotics or Good bacteria keep you healthy by supporting your immune function and controlling inflammation. Certain types of good bacteria can also:

- Help your body digest food.
- Keep bad bacteria from getting out of control and making you sick.
- Create vitamins.
- Help support the cells that line your gut to prevent bad bacteria that you may have consumed (through food or drinks) from entering your blood.
- Breakdown and absorb medications.

84. Differentiate Between Transcriptomics and Proteomic? 5 Marks

Transcriptomics

The presence of a gene in a genome does not obviously mean that it is expressed to produce mRNA and subsequently a protein. Furthermore, regulation of metabolism often involves alteration of expression of various genes, and therefore it is important to know their expression level in order to understand the physiology of the cell. Before the genome sequences became available, gene expression had to be examined on a one-by-one basis.

The techniques used included Northern blot, which involves annealing of fractionated mRNA with the cognate pieces of DNA, and reverse transcriptase– PCR, in which mRNA is used as a template to produce cDNA, which is then amplified by PCR. These methods are still used when we already know the mechanism of regulation in precise detail and when we want to know the expression level of the genes involved. However, when the regulation mechanism is not known entirely, examining the expression levels of only a few selected genes out of many thousands becomes pure guesswork and has led to many wrong conclusions.

Proteomic

Proteomic analysis requires first the separation of thousands of proteins or their fragments and then their identification. It is impossible to identify this many proteins and their fragments by using the traditional approaches, such as reactivity with specific antibodies or functional properties. Thus, the advances in mass spectrometry, now almost the universal approach in identification, were the major factor that made proteomics possible.

Previously, mass spectrometry, which measures the mass-to-charge ratio of ions in a vacuum required the “hard” ionization methods that fragmented the sample molecule. Mass spectrometry became extremely useful in the study of proteins and peptides when the two “soft ionization” methods – matrix- assisted laser desorption/ionization (MALDI) and electrospray ionization (ESI) were developed. These methods made possible the analysis of (unfragmented) peptides and proteins at high sensitivity and great precision. The significance of these advances may also be seen from the fact that two of the Nobel prizes in chemistry in 2002 were awarded to scientists who contributed to the development of these methods.

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85. Microbial biotechnology in health issues

They also have immense potential in the field of medicine. They are used industrially for the production of antibiotics, vaccines. Insulin, growth hormones and diagnostic kits. E.g. E. coli, Polio virus. Microbial development in the field of medical microbiology has helped in the protection an improvement of the health of the entire population through the development of various antibiotics. It has also helped aware the peoples regarding their health condition through the public education process. High-quality health services that lead to good physical and mental health condition is only possible through the education of public health or medical microbiology

86. Advantages of microbial biotechnology

Microbial biotechnology is a well-established branch of biotechnology, which include the application of microorganisms with emerging modern techniques of biotechnology for the development of sustainable agriculture. Microbial biotechnology deals with the manipulation through genetic engineering of living organisms or their components to produce valuable products for various applications.

Classical agriculture farming equipment and practices are reaching their limits of effectiveness in increasing agricultural productivity. The chemical fertilizers, pesticides, herbicides, and other inputs have increased agricultural production, but simultaneously, cause adverse effects on soil productivity and environmental quality. The broad application of microbes in sustainable agriculture is due to the genetic dependency of plants on the beneficial functions provided by symbiotic cohabitants. Therefore, microbial biotechnology and its applications in development of sustainable agriculture and environmental health are attracting the attention of microbiologists and biotechnologists.

India's large population is placing more pressure on natural resources like soil and water, and affecting our ability to produce sufficient food, feed, and fiber. Recently developed biotechnologies can provide appropriate new tools to find out the solutions of different specific problems of sustainable agriculture (Singh and Gupta, 2018). The aim of this study was to conclude the diversified useful applications of microbial biotechnology for the development of sustainable agriculture using different tools and techniques such as biofertilizers, biopesticides, and value additions in crops

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Past papers

1. **Define Microbial Mining? 2 Marks**

The use of microorganisms to remove minerals particularly metals from rocks. Biomining is a technique of extracting metals from ores and other solid materials typically using prokaryotes or fungi.

2. **Define probiotic? 2 Marks**

In recent years, probiotic cultures have become popular in dairy products because of their health benefits. These cultures are all very carefully selected strains, and there is good evidence that they help improve digestion, safeguard the immune system, and keep the body's intestinal flora in balance.

3. What are the criteria for Microorganism to include in probiotic group? 3 Marks

These cultures are all very carefully selected strains, and there is good evidence that they help improve digestion, safeguard the immune system, and keep the body's intestinal flora in balance.

4. Write down two Types of bio fertilizers? 5 Marks

Nitrogen bio fertilizers

- The sort of bio fertilizers can help any farmers to determine the nitrogen point during the solid ground. Nitrogen may be a crucial factor which happens to be put to use in any growing for put.
- Including, Azotobacteria must be used to your non-legume bounty; Rhizobium it takes to your legume bounty. Purple earth-friendly algae are needed to progress hemp despite the fact that Acetobacter must be used to progress sugarcane.

Phosphorus bio fertilizers

Phosphorus bio fertilizers widely used to determine the phosphorus point during the solid ground. The decision for phosphorus to your put growing is small. Phosphorus bio fertilizers come up with any solid ground obtain demanded sum of phosphorus.

- Isn't crucial that your particular selected phosphorus bio fertilizers must be used for that selected types of head. They are put to use in all different kinds of any head including; Acetobacter, Rhizobium can implement.

5. What are antimicrobial pesticides? 5 Marks

Antimicrobial pesticides

Antimicrobial pesticides help to slow the growth of human pathogenic microorganisms and are used in food processing plants, dairies, breweries, poultry houses and other animal feeding operations, hospitals and medical and dental clinics and offices, municipal drinking water and water treatment facilities, swimming pools and spas, schools, day-care centers, public access facilities and homes.

According to the U.S. Environmental Protection Agency (EPA), and defined by the Federal Insecticide, Fungicide, and Rodenticide Act, antimicrobial pesticides are used in order to control growth of microbes through disinfection, sanitation, or reduction of development and to protect inanimate objects, industrial processes or systems, surfaces, water, or other chemical substances from contamination, fouling, or deterioration caused by bacteria, viruses, fungi, protozoa, algae, or slime.

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The EPA monitors products, such as disinfectants/sanitizers for use in hospitals or homes, in order to ascertain efficacy. Products that are meant for public health are therefore under this monitoring system, including products used for drinking water, swimming pools, food sanitation, and other environmental surfaces. These pesticide products are registered under the premise that, when used properly, they do not demonstrate unreasonable side effects to humans or the environment. Even once certain products are on the market, the EPA continues to monitor and evaluate them to make sure they maintain efficacy in protecting public health.

6. Application methods of bio fertilizers? 2 Marks

- Seedling root dip,
- This method is applied to the rice crop. A bed of water is spread on the land where the crop has to grow. The seedlings of rice are planted in the water and kept there for eight to ten hours.

7. How we obtained biofuel? 2 Marks

Biofuels are fuels produced directly or indirectly from organic material biomass, including plant materials and animal waste.

8. How we apply Biofertilizer? 5 Marks

Seed treatment

Within this process your nitrogen along with phosphorus fertilizers are generally put together jointly throughout normal water. And then vegetables are generally dropped within this mix. Following purposes on this substance on the vegetables, vegetables are generally dried up. As soon as they normally dry out, they must always be sown immediately ahead of that they find harmed by simply unsafe bacteria.

Soil treatment

All the bio fertilizers along with the compost fertilizers are mixed together. They are kept for one night. Then the next day the mixture is spread on the soil where seeds have to be sown.

9. Differentiate between Submerged Fermentation and Solid-state fermentation? 5 Marks

Submerged Fermentations:

Submerged Fermentations are those in which the nutrient substratum is liquid and the organism grows inside the substratum. The culture conditions are made uniform with the help of spargers and impeller blades. Most of the industrial fermentations are of this type. The substratum which is in a liquid state and such medium is also called as broth.

Solid Substrate/State Fermentation:

Solid state (substratum) fermentation (SSF) is generally defined as the growth of the microorganism on moist solid materials in the absence or near the absence of free water. In

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recent years SSF has shown much promise in the development of several bioprocesses and products, SSF has been ambiguously used as solid-state fermentation or solid-substrate fermentation.

10. Xanthan gum. 2 Marks

Xanthan gum was discovered in the mid-1950s, during a systematic search for useful biopolymers. A bacterium Xanthomonas campestris (originally isolated from the rutabaga plant) produced a polysaccharide with potentially valuable physical properties.

11. Interaction between Plants and Microbes? 5 Marks

In nature: microbe plant interaction

- Many microbes are fond in nature and help plants to grow
- Rhizobacteria found in the soil fixate nitrogen which is required to grow many crops

Recycling of masses

Many microbes are fond in nature and help plants to grow

Rhizobacteria found in the soil fixate nitrogen which is required to grow many crops. Furthermore, microorganisms—the cyanobacteria or their DNA in the chloroplasts in plant cells—were the source of most of the free oxygen in the early atmosphere. They also oxidize ammonia (the universal end product of protein metabolism) to nitrate, which is the only nitrogen source used by plants and is therefore essential for production of our plant foods.

Microorganisms also are responsible for cellulose hydrolysis in the rumens (first stomach compartments) of cattle, facilitating the production of animal protein for human consumption.

Oxygen production

Cyanobacteria or blue green algae produce oxygen in the ocean and this oxygen production is very necessary for aquatic life survival. Blue-green algae (cyanobacteria) are prokaryotes (that is, their cells have no distinct nucleus). They are very independent nutritionally since they can perform photosynthesis using chlorophyll a. Thus, they can synthesize sugars for energy from carbon dioxide using the sun's radiation. They also release oxygen. They can respire aerobically and can fix nitrogen, generating amino acids and protein. They require only water, nitrogen gas, oxygen, carbon dioxide, some minerals, and sunlight. The evidence is that they were on earth 3.2 billion years ago. The cyanobacteria are among the earliest microorganisms and very important even today.

Decomposition

Decomposition – Defined as the breakdown of raw organic materials to a finished compost – The fungi invade the organic matter in soils first and are then followed by bacteria. – Without this recycling of inorganic nutrients, primary productivity on the globe would stop.

12. Transgenic Plant? 5 Marks

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Methods dependent on microbial biotechnology greatly increase the diversity of genes that can be incorporated into crop plants and dramatically shorten the time required for the production of new varieties of plants. It is now possible to transfer foreign genes into plant cells. Transgenic plants that are viable and fertile can be regenerated from these transformed cells, and the genes that have been introduced into these transgenic plants are as stable as other genes in the plant nuclei and show a normal pattern of inheritance.

Transgenic plants are most commonly generated by exploiting a plasmid vector carried by Agrobacterium tumefaciens, a bacterium. Foreign DNA carrying from one to 50 genes can be introduced into plants in this manner, with the donor DNA originating from different plant species, animal cells, or microorganisms. Higher plants have genes whose expression shows precise temporal and spatial regulation in various parts of plants for example, leaves, floral organs, and seeds that appear at specific times during plant development and/or at specific locations, or whose expression is regulated by light.

Other plant genes respond to different stimuli, such as plant hormones, nutrients, lack of oxygen (anaerobiosis), heat shock, and wounding. It is therefore possible to insert the control sequence(s) from such genes into transgenic plants to confine the expression of foreign genes to specific organelles or tissues and to determine the initiation and duration of such expression.

Microorganisms that live on or within plants can be manipulated to control insect pests and fungal disease or to establish new symbioses, such as those between nitrogen-fixing bacteria and plants.

13. What is a Vaccine Adjuvant? 3 Marks

An adjuvant is a substance that is formulated as part of a vaccine to enhance its ability to induce protection against infection. The word “adjuvant” comes from the Latin adjuvare and means “to help.” Adjuvants help activate the immune system, allowing the antigens—pathogen components that elicit an immune response—in vaccines to induce long-term protective immunity.

14. Give the economical beneficial of microbial biotechnology in Pakistan? 3 Marks

The Green Revolution ushered in the late 1960s has transformed some countries from a food-grain importer to a self-sufficient one. However, with our burgeoning population having already crossed the 150 million marks, the achievements of the Green Revolution are unlikely to be sufficient. Injurious but common farm practices have damaged the cultivated land through water and wind erosion, compaction, salinization, and water logging. Our forest cover is a sad mismatch of desirable levels and actual ground realities. Over-tillage rapidly depletes prime agricultural lands and today little quality land is available to increase the area under farm production. It is therefore not surprising to learn that Pakistan will need to import an estimated 10-12 million tons of food-grains to meet its basic requirement of food by the year 2050.

15. Name the adjuvants used in modern vaccine? 2 Marks

Al(OH)3 – aluminium hydroxide and/or AlPO4 – aluminium Phosphate

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16. Two Benefits of Probiotic? 2 Marks

They help improve digestion, safeguard the immune system, and keep the body's intestinal flora in balance.

17. Matrix-assisted laser desorption? 2 Marks

Soft ionization" methods – matrix-assisted laser desorption/ionization (MALDI) were developed. These methods made possible the analysis of (unfragmented) peptides and proteins at high sensitivity and great precision.

18. Is this possible to mark protein array? 2 Marks

Indeed, this is possible. In one study, 5800 open reading frames in the yeast genome were cloned and expressed as a fusion with GST and a hexa histidine tag. The proteins were purified on glutathione-agarose and attached to glass slides coated with nickel, using the affinity of the hexa histidine tag.

19. How GMOs are economically beneficial? 2 Marks

GMOs are an effective way to provide farmers a larger profit while spending less time resources. Economically they are beneficial because they are used to repel pests, which prevent the need for pesticides to be used.

20. Define Alcoholic fermentation. 3MARKS

It is done by

- Yeast and certain bacteria
- Pyruvate is broken into alcohol and carbon dioxide
- Used in the production of beer, wine and bread

21. Define the terms Antimicrobial, Antifungals. Antivirals, Anti parasitics? (3)

Antimicrobial products kill or slow the spread of microorganisms. Microorganisms include bacteria, viruses, protozoans, and fungi such as mold and mildew.

Antibacterials:

- Antibacterials are used to treat bacterial infections
- The drug toxicity to humans and other animals from antibacterials is generally considered low
- Antibacterials are among the most commonly used drugs and among the drugs commonly misused by physicians, for example, in viral respiratory tract infectionsly low.

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- Prolonged use of certain antibacterials can decrease the number of gut flora which may have a negative impacton health

Antifungals:

- Antifungals are used to kill or prevent further growth of fungi.
- In medicine, they are used as a treatment for infections such as athlete's foot, ringworm and thrush and work by exploiting differences between mammalian and fungal cells.
- They kill off the fungal organism without dangerous effects on the host.

Antivirals:

- Antiviral drugs are a class of medication used specifically for treating viral infections.
- Like antibiotics, specific antivirals are used for specific viruses.
- They are relatively harmless to the host and therefore can be used to treat infections.

Antiparasitics:

- Antiparasitics are a class of medications indicated for the treatment of infection by parasites, such as nematodes, cestodes, trematodes, infectious protozoa, and amoebae.
- Like all antimicrobials against intracellular microbes, they must kill the infecting pest without serious damage to the host

22. Describe Subunits Vaccines and give example. 5 MARKS

- Instead of the entire microbe, subunit vaccines include only the antigens that best stimulate the immune system.
- In some cases, these vaccines use epitopes the very specific parts of the antigen that antibodies or T cells recognize and bind to immunization.
- Because subunit vaccines contain only the essential antigens and not all the other molecules that make up themicrobe.

Example: Plague immunization.

23. Write down the advantages of biopesticides. (5)

- Inherently less harmful and less environmental load,
- Designed to affect only one specific pest or, in some cases, a few target organisms,
- Often effective in very small quantities and often decompose quickly, thereby resulting in lower exposures and largely avoiding the pollution problems.
- When used as a component of Integrated Pest Management (IPM) programs, biopesticides can contributegreatly.

24. Describe in detail the importance of microbial biotechnology. (5)

- Biotechnology is generally defined as application of living systems based technologies to develop commercial processes and products.

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- Over the last few decades, several fundamental discoveries in life sciences have given rise to Modern Biotechnology which is now one of the fastest growing areas of science; hence this century has rightly been termed as 'Century of Biology'
- Agricultural biotechnology offers efficient and cost-effective means to produce a diverse array of novel, value- added products and tools.
- It has the potential to increase food production, reduce the dependency of agriculture on chemicals, and lower the cost of raw materials, all in an environmentally friendly manner.

25. Explain Xanthomonas a plant pathogen. 10 MARKS

Xanthan gum was discovered in the mid-1950s, during a systematic search for useful biopolymers. In a screen of its large microbial culture collection, the Northern Utilization Research and Development Division of the U.S.Department of Agriculture discovered that the bacterium Xanthomonas campestris (originally isolated fromthe rutabaga plant) produced a polysaccharide with potentially valuable physical properties. Substantial commercial production of xanthan gum began in 1964, and in 1969 the U.S. FDA authorized its use in food.

Today, xanthan gum has numerous uses, in the food industry and elsewhere.The total U.S. consumption of xanthan in 2005 is estimated at more than 80 million pounds.

The genus Xanthomonas belongs to the γ subdivision of the proteobacteria and consists exclusively of plant pathogens. It is a relative of the genus Pseudomonas, a group of bacteria discussed elsewhere in this book in the context of the degradation of xenobiotics.

X. campestris pathovar (abbreviated "pv.") campestris, the producer of xanthan gum, causes black rot, one of the most serious diseases of plants in the genus Brassica, which includes vegetables such as cabbage, cauliflower, brussels sprouts, broccoli, rutabaga, and turnips. Seeds contaminated with bacteria spread the infection. As the seedling emerges and grows, the bacteria colonize the surface of the plant.

In the case of X. campestris pv. campestris, the epiphytic bacteria enter the internal tissues through the hydathodes, structures on the leaf margin that allow the plant to excrete water. They then migrate through the vascular system and progressively cause chlorosis (destruction of chlorophyll), vein blackening (deposition of melaninlike pigments), and, ultimately, rotting.

26. First generation Biofuels 2

First-generation biofuels are made from sugar, starch, vegetable oil, or animal fats using conventional technology. These are generally produced from grains high in sugar or starch fermented into bioethanol.

27. Pharmaceutical products 3

Many pharmaceutical drugs are also produced by microbes e.g. Cyclosporin A that is used as an immunosuppressive agent in organ-transplant patients is produced by the fungus Trichoderma polysporum.

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Statins produced by the yeast Monascus purpureus have been commercialised as blood-cholesterol-lowering agents. It acts by competitively inhibiting the enzyme responsible for the synthesis of cholesterol.

28. What are Antimicrobial pesticides? 5 (Repeat)

Antimicrobial pesticides help to slow the growth of human pathogenic microorganisms and are used in food processing plants, dairies, breweries, poultry houses and other animal feeding operations, hospitals and medical and dental clinics and offices, municipal drinking water and water treatment facilities, swimming pools and spas, schools, day-care centers, public access facilities and homes

29. Interactions between plants and microbes 5

1. Many microbes are fond in nature and help plants to grow
2. Rhizobacteria found in the soil fixate nitrogen which is required to grow many crops

Plant-microbe interactions describe a broad range of scientific studies concerning. how microbes interact with plants at the molecular biology and molecular genetics level.

- Plants and microbes can have a variety of interactions including
- Symbiotic
- Associative
- Pathogenic,
- All of which impact plant productivity, stress tolerance and disease resistance.

SYMBIOTIC ASSOCIATIONS WITH CYANOBACTERIA

- Cyanbobacteria is broadly distributed in nature and form symbiotic relationships with many different organisms.
- Cyanobacteria enhance their survival by establishing an association with a biological partner.
- Azolla: is an aquatic fern that contains bilobed leaves attached to a stem and is found floating in freshwater.
- The cyanobacteria are found in a cavity between the ventral and dorsal epidermal layers of the leaf. Associative relation of microbes and plants

ROOT ASSOCIATIONS

- The enzyme system for nitrogen fixation is found only in the prokaryotes, and in the case of symbiotic nitrogen fixation in plants,
- There is considerable specificity between the legume symbiont and bacteria for the stable association.

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- The plant provides the carbon and energy source for the bacteria to grow, while the bacteria fix nitrogen with the production of amino acids for plant growth.

PATHOGENIC INTERACTION

Bacterial Pathogens

- The bacteria-producing diseases in plants generally display a number of hydrolytic enzymes for penetration of the plant surface, and growth in the plant is generally intercellular between the parenchyma cells.
- As the disease progresses in the plant, bacterial distribution may become systemic as a result of migration into the vascular tissue.
- In some cases, the bacteria release pectinase that hydrolyzes the plant cell walls and the plant collapses.

30. How microbes enhance and improve the oil recovery? 5

One of the major concerns facing the oil industry today is the recovery of the large percentage of oil remaining unrecovered in mature and in nearly depleted oil fields.

- Microbes can influence and improve the oil recovery process by:
- Generating gases that increase reservoir pressure and decrease oil viscosity
- Generating acids that dissolve rock, thus improving absolute permeability
- Reducing permeability in channels
- Producing bio-surfactants that decrease interfacial tension
- Reducing oil viscosity by degrading long-chain saturated hydrocarbons

31. Role of microbes in petroleum industry. 10

Petroleum-based products are the major source of energy for industry and daily life. Leaks and accidental spills occur regularly during the exploration, production, refining, transport, and storage of petroleum and petroleum products.

- Release of hydrocarbons into the environment whether accidentally or due to human activities is a main cause of water and soil pollution
- Soil contamination with hydrocarbons causes extensive damage of local system since accumulation of pollutants in animals and plant tissue may cause death or mutations. Microbial processes play an important role in industry, the protection of the natural environment.

32. Proteomics and Transcriptomics

Proteomic analysis requires first the separation of thousands of proteins or their fragments and then their identification. It is impossible to identify this many proteins and their fragments by using the traditional approaches, such as reactivity with specific antibodies or functional properties. Thus, the advances in mass spectrometry, now almost the universal approach in identification, were the major factor that made proteomics possible. Previously, mass

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spectrometry, which measures the mass-to-charge ratio of ions in a vacuum required the "hard" ionization methods that fragmented the sample molecule

33. Transcriptomics

The presence of a gene in a genome does not obviously mean that it is expressed to produce mRNA and subsequently a protein. Furthermore, regulation of metabolism often involves alteration of expression of various genes, and therefore it is important to know their expression level in order tounder stand the physiology of the cell. Before the genome sequences became available, gene expression had to be examined on a one-by-one basis. The techniques used included Northern blot, which involves annealing of fractionated mRNA with the cognate pieces of DNA, and reverse transcriptase– PCR, in which mRNA is used as a template to produce cDNA, which is then amplified by PCR.

34. Inactivated Vaccines.

• Scientists produce inactivated vaccines by killing the disease-causing microbe with chemicals, heat, or radiation. Such vaccines are more stable and safer than live vaccines. Because dead microbes can't mutate back to their disease-causing state

Example: Vaccines against influenza, polio, hepatitis A, and rabies

35. Benefits of Microbes in agriculture

• Microorganisms in Agriculture

• Microorganisms have been ruling in agriculture from the past century. They are used as biofertilizers and biopesticides.

E.g. Rhizobium, Bacillus, Azotobacter

36. Role of microbes in petroleum industry?

Petroleum-based products are the major source of energy for industry and daily life. Leaks and accidental spills occur regularly during the exploration, production, refining, transport, and storage of petroleum and petroleum products.

- Release of hydrocarbons into the environment whether accidentally or due to human activities is a main cause of water and soil pollution
- Soil contamination with hydrocarbons causes extensive damage of local system since accumulation of pollutants in animals and plant tissue may cause death or mutations
- Microbial processes play an important role in industry, the protection of the natural environment
- Several practical applications of microbial processes are listed as follows

Bioremediation

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Bioremediation is a natural process that involves the use of biological entities to neutralize the contaminated site

Microbial biodegradation is the use of bioremediation and biotransformation methods to harness the naturally occurring ability of microbial xenobiotic metabolism to degrade, transform or accumulate environmental pollutants, including hydrocarbons (e.g. oil), polychlorinated biphenyls (PCBs), polyaromatic hydrocarbons

Biosurfactants

These are microbial-synthesized surface-active substances that allow for more efficient microbial biodegradation of hydrocarbons in bioremediation processes.

• There are two ways by which biosurfactants are involved in bioremediation

1) Increase the surface area of hydrophobic water-insoluble substrates. Growth of microbes on hydrocarbons can be limited by available surface area of the water-oil interface. Emulsifiers produced by microbes can break up oil into smaller droplets, effectively increasing the available surface area.

2) Increase the bioavailability of hydrophobic water-insoluble substrates. Biosurfactants can enhance the availability of bound substrates by desorbing them from surfaces (e.g. soil) or by increasing their apparent solubility. Some biosurfactants have low critical micelle concentrations (CMCs), a property which increases the apparent solubility of hydrocarbons by sequestering hydrophobic molecules into the centres of micelles.

Microbial Enhanced Oil Recovery (MEOR)

One of the major concerns facing the oil industry today is the recovery of the large percentage of oil remaining unrecovered in mature and in nearly depleted oil fields.

Microbes can influence and improve the oil recovery process by:

- generating gases that increase reservoir pressure and decrease oil viscosity
- generating acids that dissolve rock, thus improving absolute permeability
- reducing permeability in channels
- producing bio-surfactants that decrease interfacial tension
- reducing oil viscosity by degrading long-chain saturated hydrocarbons

37. Purification

In a third step the antigen will need to be purified in order to produce a high purity/quality product.

This will be accomplished using different techniques for protein purification. For this purpose several separation steps will be carried out using the differences in for instance protein size, physico-chemical properties, binding affinity or biological activity.

38. Addition of other components

DO NOT COPY

The fourth step may include the addition of an adjuvant, which is a material that enhances the recipient's immune response to a supplied antigen. The vaccine is then formulated by adding stabilizers to prolong the storage life or preservatives to allow multi-dose vials to be used safely as needed. Due to potential incompatibilities and interactions between antigens and other ingredients, combination vaccines will be more challenging to develop. Finally, all components that constitute the final vaccine are combined and mixed uniformly in a single vial or syringe.

39. Packaging

Once the vaccine is put in recipient vessel (either a vial or a syringe), it is sealed with sterile stoppers. All the processes described above will have to comply with the standards defined for Good Manufacturing Practices that will involve several quality controls and an adequate infrastructure and separation of activities to avoid cross-contamination, as shown in the diagram below. Finally, the vaccine is labelled and distributed worldwide.

40. Charging electronics?

Charging Electronics • According to scientists from Saint Luis University, a fuel cell was developed with cooking oil and sugar to generate electricity

• consumers will be able to use these cells instead of generating electricity. • Consumers may be able to use fuel cells in place of batteries to charge anything from computers to cell phones

41. Nitrogen bio fertilizers?

• The sort of bio fertilizers can help any farmers to determine the nitrogen point during the solid ground. Nitrogen may be a crucial factor which happens to be put to use in any growing for put.

• Including, Azotobacteria must be used to your non-legume bounty; Rhizobium it takes to your legume bounty. Purple earth-friendly algae are needed to progress hemp despite the fact that Acetobacter must be used to progress sugarcane

42. Steroids

• Steroids are a very important group of chemicals, which are used as anti-inflammatory drugs, and as hormones such as estrogens and progesterone, which are used in oral contraceptives.

• Producing steroids from animal sources or chemically systhesising them is difficult, but microorganisms can synthesise steroids from sterols or from related compounds

43. Worm composting?

Worm composting is using worms to recycle food scraps and other organic material into a valuable soil amendment called vermi compost, or worm compost.

44. Define compost (Repeat)

DO NOT COPY

Compost is organic matter that has been decomposed and recycled as a fertilizer and soil amendment

• Compost is a key ingredient in organic farming.

• End product of the decomposition of organic matter

Composting

• Composting is nature's process of recycling decomposed organic materials into a rich soil known as compost

• Composting is a natural process that turns organic material into a dark rich substance

45. What is biodiesel and how it is produced 3 marks (Repeat)

Biodiesel is produced, mainly in the European Union, by combining vegetable oil or animal fat with an alcohol. Biodiesel can be blended with traditional diesel fuel or burned in its pure form in compression ignition engines. Its energy content is somewhat less than that of diesel (88 to 95%).

Biodiesel can be derived from a wide range of oils, including rapeseed, soybean, palm,coconut or jatropha oils and therefore the resulting fuels can display a greater variety ofphysical properties than ethanol.

46. Name steps of vaccine purification 5 marks

Steps of vaccine production

1. Generation of the Antigen VIRUS BACTERIA
2. Isolation of the Antigen
3. Purification
4. Addition of Other Components
5. Packaging

Generation of the Antigen VIRUS BACTERIA:

- The first step in order to produce a vaccine is generating the antigen that will trigger the immune response. For this purpose the pathogen's proteins or DNA need to be grown and harvested using the following mechanisms:
- Viruses are grown on primary cells such as cells from chicken embryos or using fertilised eggs (e.g. influenza vaccine) or cell lines that reproduce repeatedly (e.g. hepatitis A)
- Bacteria are grown in bioreactors which are devices that use a particular growth medium that optimises the production of the antigens

DO NOT COPY

- Recombinant proteins derived from the pathogen can be generated either in yeast, bacteria or cell cultures.

Isolation of the Antigen:

- The aim of this second step is to release as much virus or bacteria as possible.
- To achieve this, the antigen will be separated from the cells and isolated from the proteins and other parts of the growth medium that are still present.

Purification:

- In a third step the antigen will need to be purified in order to produce a high purity/quality product.
- This will be accomplished using different techniques for protein purification. For this purpose several separation steps will be carried out using the differences in for instance protein size, physico-chemical properties, binding affinity or biological activity.

Addition of Other Components:

- The fourth step may include the addition of an adjuvant, which is a material that enhances the recipient's immune response to a supplied antigen. The vaccine is then formulated by adding stabilizers to prolong the storage life or preservatives to allow multi-dose vials to be used safely as needed.
- Due to potential incompatibilities and interactions between antigens and other ingredients, combination vaccines will be more challenging to develop. Finally, all components that constitute the final vaccine are combined and mixed uniformly in a single vial or syringe.

47. Role of antimicrobial pesticides (5)

Antimicrobial pesticides help to slow the growth of human pathogenic microorganisms and are used in food processing plants, dairies, breweries, poultry houses and other animal feeding operations, hospitals and medical and dental clinics and offices, municipal drinking water and water treatment facilities, swimming pools and spas, schools, day-care centers, public access facilities and homes

48. Challenges in adjuvant (5)

Challegnes with adjuvant

- Vaccines containing novel adjuvant formulations are increasingly reaching advanced development and licensing stages, providing new tools to fill previously unmet clinical needs.

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- However, many adjuvants fail during product development owing to factors such as manufacturability, stability, lack of effectiveness, unacceptable levels of tolerability or safety concerns.
- Lower reactogenicity adjuvants came with reduced vaccine efficacy.
- Safety and tolerability are critical regulatory issues facing the adjuvants.
- In most cases increased adjuvant potency is associated with increased reactogenicity and toxicity.

49. Which type of relationship between plants and microbes

Plant-microbe interactions

Plant-microbe interactions describe a broad range of scientific studies concerning how microbes interact with plants at the molecular biology and molecular genetics level.

Plants and microbes can have a variety of interactions including

Symbiotic, Associative, Pathogenic, all of which impact plant productivity, stress tolerance and disease resistance.

Root Associations

- The enzyme system for nitrogen fixation is found only in the prokaryotes, and in the case of symbiotic nitrogen fixation in plants,
- There is considerable specificity between the legume symbiont and bacteria for the stable association.
- The plant provides the carbon and energy source for the bacteria to grow, while the bacteria fix nitrogen with the production of amino acids for plant growth.

Pathogenic interactions

• Bacterial Pathogens

• The bacteria-producing diseases in plants generally display a number of hydrolyticenzymes for penetration of the plant surface, and growth in the plant is generally intercellular between the parenchyma cells. ›

• As the disease progresses in the plant, bacterial distribution may become systemic as a result of migration into the vascular tissue.

• In some cases, the bacteria release pectinase that hydrolyzes the plant cell walls and the plant collapses.

50. Role of microbes in oil recovery process (5)

• One of the major concerns facing the oil industry today is the recovery of the large percentage of oil remaining unrecovered in mature and in nearly depleted oil fields.

• Microbes can influence and improve the oil recovery process by:

DO NOT COPY

• generating gases that increase reservoir pressure and decrease oil viscosity

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• reducing permeability in channels

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• reducing oil viscosity by degrading long-chain saturated hydrocarbons

51. Main object in transgenic plants (5)

The main objectives of creating transgenic plants are attempts to engineer metabolic pathways for the production of tailor-made plant polymers or low molecular weight compounds, increased resistance towards pathogens and pesticides, improved food quality, and the production of polypeptides for pharmaceutical or technical use.

52. Benefits of compost

Composting Benefits

• Soil conditioner: With compost, you are creating rich humus for lawn and garden. This adds nutrients to your plants and helps retain moisture in the soil.

• Recycles kitchen and yard waste: Composting can divert as much as 30% of household waste away from the garbage can.

• Introduces beneficial organisms to the soil: Microscopic organisms in compost help aerate the soil, break down organic material for plant use and ward off plant disease.

• Good for the environment: Composting offers a natural alternative to chemical fertilizers.

• Reduces landfill waste: Most landfills in North America are quickly filling up; many have already closed down. One-third of landfill waste is made up of compostable materials

53. Nitrogen bio fertilizers?

• The sort of bio fertilizers can help any farmers to determine the nitrogen point during the solid ground. Nitrogen may be a crucial factor which happens to be put to use in any growing for put.

• Including, Azotobacteria must be used to your non-legume bounty; Rhizobium it takes to your legume bounty. Purple earth-friendly algae are needed to progress hemp despite the fact that Acetobacter must be used to progress sugarcane

54. Steroids ?

• Steroids are a very important group of chemicals, which are used as anti-inflammatory drugs, and as hormones such as estrogens and progesterone, which are used in oral contraceptives.

• Producing steroids from animal sources or chemically systhesising them is difficult, but microorganisms can synthesise steroids from sterols or from related compounds

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55. Steps in transgeneic plants?

Genetic modification involves the mutation, insertion, or deletion of genes.

• Inserted genes usually come from a different species in a form of horizontal gene transfer

• In nature this can occur when exogenous DNA penetrates the cell membrane for any

Reason

This can be accomplished artificially by

- attaching the genes to a virus.
- physically inserting the extra DNA into the nucleus of the intended host with a very small syringe.
- using electroporation (that is, introducing DNA from one organism into the cell of another by use of an electric pulse).
- firing small particles from a gene gun

56. What are the physical factors of microbial environment?2 mark

- Physical factors
- Temperature
- Humidity
- Ph
- viscosity

57. What are the physiological factors of strain production?3 mark

Physiological Factors:

- Morphology
- Metabolic by-products
- Shear Sensitivity
- Genetic Stability
- Issues in production strain:

58. Write two names of pharmaceutical drugs produced by microbs and how these are produced?5

- Many pharmaceutical drugs are also produced by microbes e.g. Cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus Trichoderma polysporum.
- Statins produced by the yeast Monascus purpureus have been commercialised as blood-cholesterol-lowering agents. It acts by competitively inhibiting the enzyme responsible for the synthesis of cholesterol.

59. Name the vatimns produced by microbs? 3 mark

DO NOT COPY

- Vitamins are some organic compounds which are capable of performing many life-sustaining functions inside our body.
- These compounds cannot be synthesised by humans, and therefore, they have to be supplied in small amounts in the diet.
- Microbes are capable of synthesizing these compounds and hence they can be used for the commercial production of many of the Vitamins e.g. thiamine (Vitamin B1), riboflavin (Vitamin B2)

60. Role of microbes in Meat industry? (3 marks)

Meat starter cultures are used to make dried, fermented products

- such as salami, Salami is a type of cured sausage consisting of fermented and air-dried
- meat, typically beef or pork.
- pepperoni, Pepperoni is an American variety of salami, made from cured pork and beef
- mixed together and seasoned with paprika or other chili pepper.
- dried ham, Ham is pork that has been preserved through salting, smoking, or wet curing
- Lactic bacteria develop the flavour and colour of the products.
- In addition, a wide variety of moulds are used to ripen the surface of sausages, preserving the natural quality of the product and controlling the development of flavour

61. name physiological effects in strain?

Physiological Factors

- Morphology
- Metabolic by-products
- Shear Sensitivity
- Genetic Stability
- Issues in production strain:

62. Alcholic fermentation?

Done by

• Yeast and certain bacteria

• Pyruvate is broken into alcohol and carbon dioxide

• Used in the production of beer, wine and bread

DO NOT COPY

63. Bio processing of microbes

Application of natural or genetically manipulated whole cells/ tissues/ organs, or parts thereof, for the production of industrially or medically important products. Encompasses research, development, manufacturing, and commercialization of products prepared by using biological systems

• Issues of microorganisms in bio processing

1st Step:

• Isolation of Pure or Mixed Cultures from nature

2nd Step:

Assesment to determine the capacity for the production

Issues of microorganisms in bio processing.

64. microbs are beneficial for human body3

In recent years, probiotic cultures have become popular in dairy products because of their health benefits.

• These cultures are all very carefully selected strains, and there is good evidence that they help improve digestion, safeguard the immune system, and keep the body's intestinal flora in balance.

65. How animals digest cellulose.

IN DIGESTION

• ONE OF THE MAIN COMPONENT OF PLANT IS CELLULOSe.

Cellulose is hard to digest by the animal…

• How they digest the cellulose…?

• Cellulose can be digest by enzyme named cellulase.

• How they can get cellulase?

• Herbivores depend on bacteria that live in their stomach to digest cellulose

• This bacteria secretes (merembeskan) cellulase into the stomach

66. Use of Microbes in Medicine

MEDICINE

• Some type of microorganism produce important drugs called ANTIBIOTICS.

• Functions of antibiotics….

DO NOT COPY

• Weaken, Destroy other microorganism

• Penicillium notatumis a GREEN MOULD

• Its produces antibiotic called PENICILLIN

67. Psychological Factors in Strain improvement.

Physiological Factors

• Morphology

• Metabolic by-products

• Shear Sensitivity

• Genetic Stability

• Issues in production strain:

68. Importance of Microbial Genomics and microbial Biotechnology.

Microbial genomics and microbial biotechnology research is critical for advances in food safety, food security, biotechnology, value-added products, human nutrition and functional foods, plant and animal protection, and furthering fundamental research in the agricultural sciences. Microbial biotechnology, enabled by genome studies, will lead to breakthroughs such as improved vaccines and better disease-diagnostic tools, improved microbial agents for biological control of plant and animal pests, modifications of plant and animal pathogens for reduced virulence, development of new industrial catalysts and fermentation organisms, and development of new microbial agents for bioremediation of soil and water contaminated by agricultural runoff.

Application of scientific and engineering principles to the processing of materials by microorganisms to create useful products or processes

• Microorganisms utilized may be natural isolates, laboratory selected mutants or microbes that have been genetically modified using recombinant DNA methods

• Deals with the prevention of deterioration of processed or manufactured good, environmental protection and waste disposal system

• Production of antibiotics, organic acids and enzymes by fermentation of microbes

69. Name Pharmaceutical Drugs produced by microbes.

Pharmaceutical Drugs

• Many pharmaceutical drugs are also produced by microbes e.g. Cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus Trichoderma polysporum.

DO NOT COPY

• Statins produced by the yeast Monascus purpureus have been commercialised as bloodcholesterol-lowering agents. It acts by competitively inhibiting the enzyme responsible for the synthesis of cholesterol.

70. Why plants need nitrates? 2 (Repeat)

AGRICULTURE

• Bacteria play important roles to promote the plant growth.

• Bacteria, Break down dead plants and dead animals by releasing AMMONIA

• AMMONIA, Adds nitrogen into the soil

• The nitrogen fixing bacteria in the soil change the ammonia into nitrates.

• This nitrates will be used by plants

71. How microbe are important inside human body ? 3

In our bodies, different microbes help to digest food, ward off invasive organisms, and engage in skirmishes and pitched battles with the human immune system in the give-andtake of the natural disease process.

72. How microbes are important in industry (2)

Dairy industry

• It would be impossible to make cheese without a starter culture.

The production of cheese utilizes starter cultures being added to milk with the aim of producing semi-hard and hard cheeses. The starter cultures produce lactic acid, resulting in a pH decrease in the milk and promoting curd synthesis

• Yogurt is also a fermented milk product, which takes advantage of lactic acid generating bacteria cultures Lactobacillus and their use of lactose in milk as a substrate to produce lactic acid

• In yoghurt and other fermented milk products, the culture is responsible for the taste and texture of the final product.

• In recent years, probiotic cultures have become popular in dairy products because of their health benefits.

• These cultures are all very carefully selected strains, and there is good evidence that they help improve digestion, safeguard the immune system, and keep the body's intestinal flora in balance.

73. What is alcoholic fermentation (3)

Lactic acid fermentation

DO NOT COPY

• occurs in animal muscles when the tissue requires energy at a faster rate than oxygen can be supplied

• used to convert lactose into lactic acid in yogurt and cheese production

• also gives the sour taste to fermented vegetables such as pickles

74. Role of microorganisms in industries

Microbes are widely used in large-scale industrial processes microbes are used to synthesise a number of products valuable to human beings.

• This Industry has provided products that have profoundly changed our lives and lifespans.

• These products include beverages, food additives, products for human and animal health,and biofuels.

Beverages

• Microbes especially yeast have been used from time immemorial for the production ofbeverages like Wine, Beer, Whisky, Brandyor Rum.

• For this purpose, the yeast Saccharomyces cerevisiae(commonly called Brewer's yeast) is used for fermenting malted cereals and fruit juices to produce ethanol.

Antibiotics

• Antibiotics produced by microbes are regarded was one of the most significant discoveries of the twentieth century and have made major contributions towards the welfare of human society.

• Many antibiotics are produced by microorganisms, predominantly by Actinomycetes in the genus Streptomycin (e.g. Tetracycline, Streptomycin, Actinomycin D) and by filamentous fungi (e.g. Penicillin, Cephalosporin).

Organic acids

• Microbes are also used for the commercial and industrial production of certain organic acids. These compounds can be produced directly from glucose (e.g. gluconic acid) or formed as end products from pyruvate or ethanol.

• Examples of acids producers are Aspergillus Niger (a fungus) of Citric acid, Acetobacter acute (a bacterium) of acetic acid, Lactobacillus (a bacterium) of lactic acid and many others.

Enzymes

• Many microbes synthesise and excrete large quantities of enzymes into the surrounding medium. Using this feature of these tiny organisms, many enzymes have been produced commercially.

These include Amylase, Cellulase, Protease, Lipase, Pectinase, Streptokinase and many others.

DO NOT COPY

• Enzymes are extensively used in food processing and preservation, washing powders, Leather Industry, Paper Industry and in scientific research

Vitamins

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Pharmaceutical Drugs

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